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Named Entity Recognition in German-language Telegram channels

Bachelor Thesis

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Faculty 4

by

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Abstract

This thesis implements Named Entity Recognition (NER) in German-language text data disseminated via the messenger service Telegram, with a focus on the mobilization for protests against measures to counteract the COVID-19 in Germany.

This work explores the potentials and limitations of two existing German-language models for NER, namely *spacy* and *hugging face*, for the recognition of the entity types PERSON, LOCATION, and ORGANIZATION.

Furthermore, it makes use of the *ctparse* module to extract text parts containing DATE and TIME information.

This thesis contributes to the task of German-language NER in general by applying it to a specific domain and data type, namely text data from messenger services. Furthermore, its aim is to provide a low-resource tool for both researchers and civil society for monitoring and analyzing large amount of text data from social media/messengers.

Dedication

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.

I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.

I am aware that the violation of this regulation will lead to the failure of the thesis.

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Chapter 1

Introduction

1.1 Background

The outbreak of the COVID-19 pandemic lead to an increased spread of misinformation and conspiracy theories evolving around e.g. the origin of the virus, but also about the intentions of the responsible authorities putting into effect countermeasures to prevent the further spread of the virus such as e.g. lockdowns, the mandatory use of face masks or testing, and last but not least COVID-19 vaccinations.

The rise of conspiracy theories has often been accompanied by anti-semitism, racism, but also hateful targeting of politicians.

- racism: especially Asian community made responsible for spreading the virus
- antisemitism: Jews identified as culprits behind the pandemic, using it as a means to obtain world dominance
- politicians: complicit with those who benefit from pandemic; following a secret plan; restriction of civil rights
- anti-COVID-19 measures were accompanied by protests against them

- freedom of speech, criticizing the government are valuable and essential components of a democratic society
- but: often hateful tendencies and developments, both online and offline
- entanglement of digital and real world
- hate does not stay in the internet, also has real effects
- examples: ... Karl Lauterbach, Kretschmer, Demo vor Haus von Politikerin, Entführung Gesundheitsminister in Schweiz
- spread of misinformation, conspiracy theories: social media and messengers
- not only information, also mobilization
- special targets: politicians, but also minority groups, e.g. persons identified as Jewish
- journalists uncover networks, raise attention for dangerous potential of these political networks, movements
- authorities react only slowly
- amount of text data make it hard for reserachers, NGOs to monitor development and mobilization strategies

1.2 Objective

- low-resource approach of NER in German-language Telegram messages
- extration of entities particularly relevant for mobilization / monitoring of potential targeting: PERSON, ORGANIZATION; DATE; TIME; LOCATION; ACTION as custom type

- explore potentials and limitations of existing models
- explore potentials of translation approach
- provide a low-resource tool for NGOs and researchers make available as API; Project teledash

1.3 Thesis Structure

- Theory
- Implementation
- Conclusion and Discussion

Chapter 2

Theory

2.1 Natural Language Processing

2.1.1 Definition

2.1.2 Historical evolvement

2.1.3 Challenges for NER

2.2 NER approaches

2.2.1 Rule-based

2.2.2 Machine-learning based

2.2.3 Neural Networks

2.2.4 Summarization

Chapter 3

Implementation

3.1 Data

3.1.1 Existing German-language datasets for NER

3.1.2 Domain-specific data

Telegram channel ‘Demotermine’: 52.900 subscribers by the beginning of June 2022,

long channel name: ‘Demo-Termine Kontakte, Gleichgesinnte treffen - Demokalender, Spaziergänge, Mahnwachen, Meditation *Rücktritt Bundesregierung!’

channel-info: “ Wir Demokraten wollen miteinander in Kontakt bleiben @DemotermineChat und uns im Bedarfsfall möglichst schnell austauschen, Beweise in Wort, Bild und Ton sichern, damit sie nicht unterschlagen werden können @LivestreamsFuerDich! @Terminverteiler_{Bot}’

TODO insert histplot to show development of message numbers

date range: 2018-10-05 bis 2021-12-19

total number of messages (wo duplicates): 27.627 messages

German-language: 23.911

Sample size: xx out of 23.911 confidence level 95

text length: min 100 (lower quartile), max 350 (upper quartile)

-i remaining:

3.2 Models

- spacy - hugging face

3.3 Pipeline

3.4 Evaluation metrics

3.4.1 Accuracy

3.4.2 Performance

3.5 Translation approach?

3.6 Summarization

Chapter 4

Results

4.1 Spacy

4.2 Hugging Face

4.3 Translation approach

4.4 Summarization

Chapter 5

Conclusion

5.1 Summarization

5.1.1 subsection

5.2 Discussion

Appendix

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